Annie Prasanna Manoharan

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Education

University of Arkansas Aug 2024 – May 2026
Master of Science in Computer Engineering *Fayetteville, AR*
Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Privacy Enhancing Techniques

Sri Ramakrishna Engineering College Aug 2019 – May 2023
Bachelor of Engineering in Electronics and Communication *India*

Technical Skills

Cloud and Tools: AWS Step Functions, Amazon API Gateway, AWS Glue, AWS Lambda, Design patterns, AWS CDK, Amazon DynamoDB, Amazon S3, Prompt engineering, Docker, Git, Linux, Data validation and quality, Vector embeddings, CI/CD, Agile/Scrum
Data Engineering: ETL/ELT, Apache Airflow, Apache Kafka, Apache Flink, MySQL, PostgreSQL
Machine Learning and AI: LangChain, RAG, Flask, Pandas
Languages: SQL, Python

Professional Experience

Data Engineering Intern Jul 2024 – Present
InsOps Inc. *Chicago, Illinois*

- Developed data pipelines in AWS using AWS Glue, Lambda, EMR, SQL, and Apache Airflow for orchestration, with S3 cloud storage, for scalable data ingestion, transformation, validation, and analytics
- Built data workflows integrating SQL Server, Azure SQL, AWS S3, and cloud databases, preparing structured datasets for downstream data science and analytics

Teaching Assistant Sep 2024 – May 2026
University of Arkansas, Dept. of Computer Science and Engineering *Fayetteville, AR*

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals

Research Assistant Sep 2024 – Jan 2026
AICV Lab, University of Arkansas *Fayetteville, AR*

- Built reusable preprocessing workflows to segment, organize, and transform raw image/video data into structured inputs for downstream processing and evaluation
- Evaluated results using mAP, MOTA, IDF1, PSNR, SSIM, and FID

Project Engineer Jun 2023 – Jun 2024
Soliton Technologies *India*

- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation
- Developed customized software solutions for Dräger as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements

Projects

Serverless Expense Reconciliation API *Python, AWS Lambda, AWS Step Functions, Amazon API Gateway, Amazon DynamoDB, Amazon S3*

- Built an event-driven reconciliation service on AWS Lambda and Step Functions that ingests expense extracts from S3, validates records against expected totals, and flags mismatches for review
- Exposed REST endpoints through Amazon API Gateway backed by Python handlers, with DynamoDB tables storing transaction state and audit history
- Defined the deployment as infrastructure as code with AWS CDK, covering IAM roles and environment configuration for repeatable deployments

Local LLM-Powered News Research Platform *Python, LangChain, Streamlit, FAISS, Ollama, Llama 3.2*

- Built an end-to-end local ETL pipeline using Streamlit and WebBaseLoader to ingest, clean, chunk, and preprocess unstructured web content on demand
- Designed a local vector store with FAISS and nomic-embed-text to serialize high-dimensional embeddings, bypassing redundant ingestion and reducing processing latency